

TrES-3 b

R-0006

Practice run · workflow W8-validation · record 2026-07-07_TrES-3b_fresh · created 2026-07-08T01:22:15+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2461219.7213 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.056
Transit depth (Rp/Rs)^2	3.84 +/- 2.19 [%]
Orbital Inclination (inc)	85.58 +/- 2.88
Airmass coefficient 1 (a1)	0.43 +/- 0.27
Airmass coefficient 2 (a2)	0.68 +/- 1.25
Scatter in the residuals of the lightcurve fit is	52.8 %
Best Comparison Star	#2 - [570, 387]
Optimal Aperture	1.32
Optimal Annulus	11.39
Transit Duration (day)	0.075 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.196 ± 0.056 vs 0.16309 (deviation 0.59σ)
Duration fit vs published	0.075 d vs 0.0567 d (deviation 1.15σ)
Mid-time O–C	-5.1 min
Depth SNR	3.5
Residual scatter	52.8 %

Light curve

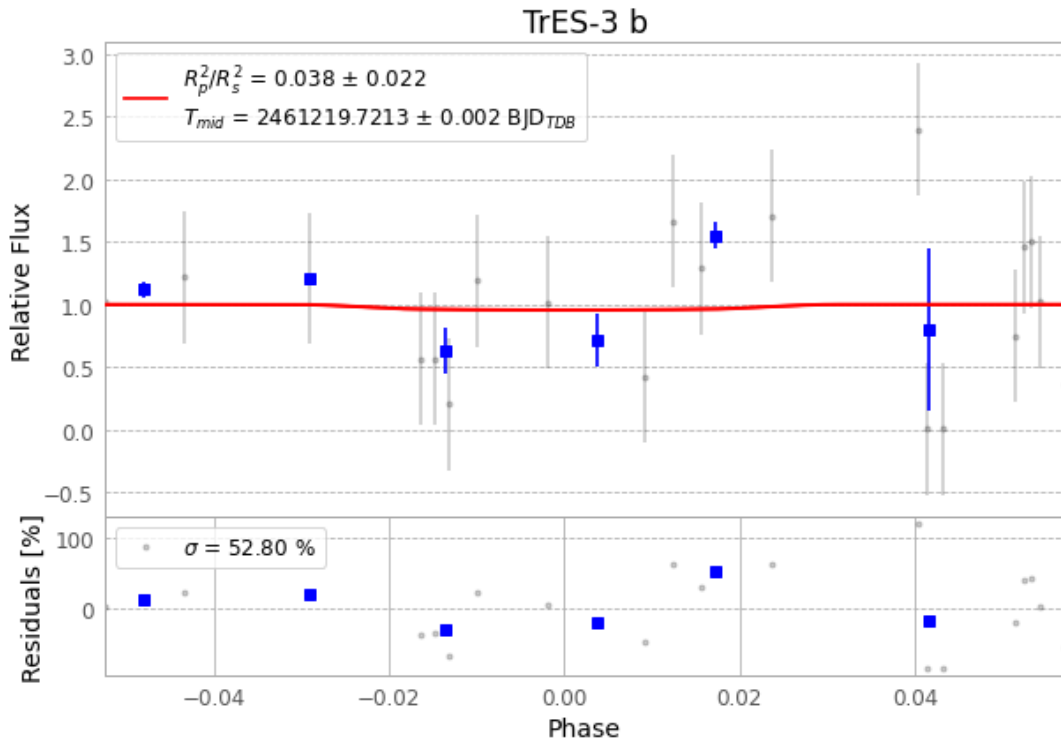


Figure 1 — FinalLightCurve_TrES-3 b_2026-06-27.png (SHA-256 11eac2dfc9f71e2f...)

Observer notes

Fresh public-archive night from MicroObservatory "Cecilia" (2026-06-28, 70 frames). The fit is unusable: 52.8% residual scatter, R_p/R_s^* uncertainty ~30%, duration inconsistent with published — consistent with clouds or no transit in the observed window. Kept as a REJECT-shaped validation case: this is exactly the dossier the human review gate exists to catch.

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [285, 211], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 49417baa7772a5c0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

70 FITS frames (46 MB), TRES-3260628033421.FITS → TRES-3260628070015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: exotic-tres3.log (c69d54a60466...), FinalLightCurve_TrES-3 b_2026-06-27.png (11eac2dfc9f7...), FinalLightCurve_TrES-3 b_2026-06-27.csv (c71924271a62...)